

UTRECHT UNIVERSITY

MASTER THESIS

**Physics-informed motion prediction for UAV
collision detection**

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Abstract

An enormous amount of the data humanity collects is visual data from monocular cameras. Despite this amount of data, there is a lack of research in the field on how the motion represented in these videos can be modelled in a physically accurate way. This thesis aims to develop a way to model optical flow data from an ordinary monocular camera with physics-informed machine learning and then to prove its effectiveness on the UAV collision detection problem. For this task, a Hamiltonian neural network is used to model optical flow measurements to predict a future trajectory of an object. These predictions are then used in a collision detection system that can work on data without any annotations. The approach is proven effective in modelling optical flow with a rigid physics-informed machine learning model. At the same time, a complex representation of the motion from several observations is proven to predict collisions accurately without sacrificing time to respond. The key takeaway from this study is that low-fidelity visual data, despite being an approximation of real-world motion, can describe fundamental physical properties in it.

Contents

1	Introduction	4
1.1	Research Context	4
1.2	Research gap	5
1.3	Research Question Formulation	5
1.4	Contributions	6
1.5	Structure	7
2	Literature Review	9
2.1	Physical Optical Flow	9
2.1.1	Kalman Filter	9
2.1.2	Temporal Coherent Optical Flow	12
2.1.3	Fluid Dynamics	12
2.1.4	Summary	14
2.2	Physics Informed Machine Learning	14
2.2.1	Hamiltonian Neural Network	16
2.2.2	Physics Informed Computer Vision	18
2.3	Collision Detection	20
2.4	ColANet Dataset	22
3	Methodology	23
3.1	Research design	23
3.2	Data preparation	24
3.2.1	Segmentation	25
3.2.2	Optical flow	26
3.2.3	K means clustering	29
3.2.4	Score for collision	30
3.3	Data modeling	31
3.3.1	Hamiltonian neural network	31
3.4	Conclusion	33
4	Results and discussion	34
4.1	Validation	34
4.2	HNN Training Loss on Optical Flow Data	36
4.3	Results with Original Score Algorithm	37
4.4	Results with Improved Score Algorithm	38
4.5	Visual example	39
4.6	Rotational motion from optical flow	41
4.7	Limitations	42
4.7.1	Data limitations	42
4.7.2	Segmentation limitations	42
4.7.3	Optical flow limitations	42
4.7.4	Lable limitations	43
4.7.5	HNN limitations	43
4.7.6	Real world limitations	43
5	Conclusion	44
5.1	Recommendations for future work	44

5.2 Take away	44
Bibliography	45

1 Introduction

1.1 Research Context

In the digital age, people have understood the value of information in our world, with all the possible insights found in it. The defining discipline that is able to gather this insight on a massive scale is machine learning. Unfortunately, due to its complexity and lack of precisely annotated sensor data, the field of engineering was slow to pick up the benefits of the machine learning revolution. Modern neural networks can find countless correlations in data, but not all of these can obey the governing laws of physics. However, with better embedding of knowledge about the laws of nature, AI models can help expand this understanding in unimaginable directions.

A novel solution to the problem of modelling data in a physically accurate way is the field of **physics-informed machine learning (PIML)** [27], which aims to embed physical properties into neural network solutions. Following this approach, models learn correlations between features as well as correlations that describe the physical properties of the data. Therefore, the predictions of the network are not just accurate, but aligned with the governing laws of nature in our world. Additionally, adding physical priors to the models reduces the required data and computation since there are fewer parameters to learn. Nevertheless, the biggest advantage of these models is that they are much more interpretable than standard neural networks, which are often accused of being a black box.

The method of adding physical priors can be done in several ways. The most common one is incorporating a second loss function, which only evaluates if the answer of the neural network (NN) satisfies physical equations, such as conservation of energy or symmetries. This is the defining feature of the most well-known architecture: physics-informed neural network (PINN) [46]. Another way is to modify the training data to explicitly focus on physical characteristics. A less common option is to impose hard constraints on the whole architecture so the results have less room for deviation from physically accurate results.

A great innovation in recent years has been the development of the Hamiltonian neural networks [25]. These models aim to integrate Hamilton’s equation for the conservation of energy in the loss function of the neural network architecture. Furthermore, they are particularly useful in modelling chaotic systems even with low-fidelity data, which signifies a big improvement to standard approaches like neural ordinary differential equation(ODE) models.

People use all their senses to gather information about the real world, and the most valued one is the eyes [16]. All this valuable visual information can be perceived by AI models due to the field of computer vision. Here, information about the world can be extracted from one of the most widely used and affordable sensors - the ordinary video camera. This representation describes the three-dimensional real world through a two-dimensional plane. Such an opportunity opens the possibility of AI models to learn from the vast amount of video data humanity collects every day.

To model one physical event, it is necessary to observe changes in an object over time that can be described with a mathematical equations. The most widely used mechanism

that the field of computer vision uses to measure the changes in the position over time is **optical flow** [51]. Optical flow measures the displacement of pixels between frames in a video. To do so, several assumptions have to be in place. The main assumption of optical flow is that the average brightness in the image does not change between frames. This guarantees that changes in colour are only caused by movements, not changes in the illumination in the image. Another assumption is that changes occur more gradually, as such a big displacement of objects is not possible. The last most commonly used assumption is that neighbouring pixels should have similar optical flow for coherent results. With all these assumptions about the movement in the image plane, it is clear that this is more of an approximation of the real motion rather than a precise measurement of a dedicated sensor. Nevertheless, because the real world is based on physical principles, its translation to the image plane will carry these principles in a transformed way.

In recent years, the technology and applications of **unmanned aerial vehicles (UAVs)** have seen a dramatic surge. Some of these UAVs use monocular cameras as their sensor of choice to navigate in the real world, due to their low cost, size, and weight. The problem of how to use sensor information to alert the UAV for crashes with the surrounding objects is called collision detection. Here, the task involves calculating the trajectory and time to contact incoming obstacles in the image plane. This is usually determined by a combination of factors like the expansion of an object or changes in its position in the image plane. To determine these changes, many solutions to this problem use optical flow for this task [23]. However, the usage of optical flow with all the assumptions makes the system prone to errors, which can have catastrophic effects when navigating in urban areas. UAVs are often used in close proximity to people, therefore, models about their navigation need to be more interpretable so potential risks can be examined easily.

1.2 Research gap

A lot of experiments with PIML in recent years have been dedicated to scientific research due to the need for domain knowledge of the physical laws present in the studies. The data from scientific simulations with well-described physical properties is only a very small amount of the data that humanity collects and is available. That leaves a significant gap in the research on modelling low-quality visual data in a physically grounded way. This is an unrealised possibility since there are fields of study, like computer vision, with lots of available data already based on physical laws, but modelled just in terms of correlations.

Currently, optical flow motion is not estimated with the ground laws of motion taken into account. It is just an approximation, based on the pattern that displacement of similarly coloured pixels indicates motion in the corresponding direction. However, even if not all changes are scientifically correct, the general movement of objects obtained with optical flow could give general insights into the motion in the real world.

1.3 Research Question Formulation

The research aim is to explore the relation between visual sensors and physical machine learning models. Based on this, the following research questions have been formed, and

the following research objectives have been outlined:

Table 1: Research Questions and Objectives

Research Question	Research Objective
RQ1 Can optical flow be transformed from an approximation of the movement to a physically accurate motion field when modelled with physics-informed machine learning?	RO1 Use a Hamiltonian neural network to model the optical flow of a segmented object at consecutive frames of a video.
RQ2 Can physically accurate optical flow improve collision detection for UAVs from visual data?	RO2 Develop a model that uses physically accurate optical flow to predict the chance of collision and validate it against appropriate models.

1.4 Contributions

The contribution of this thesis is first to develop a physics-informed machine learning model for motion prediction of objects based on their optical flow. Second, the predictions from this model are used in the collision detection model.

This thesis prepares an experiment where the motion gathered from optical flow can be modelled. As such, a collision detection system is developed on videos of UAVs crashing into objects. This system is meant to work with data only from **monocular camera**. First, the system uses video segmentation to extract the most important objects in the image into isolated segments. These segments represent an outline of an object that is tracked over the frames. After that, the optical flow for each segment is calculated. Afterwards, the collected optical flow is modelled with a Hamiltonian neural network. Finally, optical flow and predictions from the Hamiltonian neural network are used in a feedforward neural network to predict the chance of collision from a particular segment.

Taking into consideration that the trajectory of the incoming obstacle is likely to be relatively stable over time, the model does not consider information from more than one frame at a time. Instead, information is used on how long the segment has been present on the image plane.

The main advantage of the proposed method is the provision of a **much more explainable physics-based model**, due to the constraints in physical solutions. At the same time, this model does not rely on annotated data but can work on raw inputs.

The effectiveness of the proposed model is compared to four models:

Model Variant	Description
Baseline model	This model uses optical flow information, future predictions of the optical flow, and one parameter for the number of frames the segment was present.
Comparison model A	The same as the baseline model, but without the future predictions with the Hamiltonian network.
Comparison model B	The same as the baseline model, but incorporating a long short-term memory (LSTM) model with ten frames instead of the number of frames feature.
Comparison model C	Recurrent neural network (RNN) from ten frames without information about optical flow.
Comparison model D	A convolutional neural network (CNN) with information on the red, green, and blue (RGB) values of the pixels.

Table 2: Description of the models used to prove the research question.

The key metrics to compare the performance will be:

- Accuracy - the Hamiltonian neural network has to be able to give predictions that reflect the motion gathered from optical flow. Simultaneously, the collision detection models need to be able to differentiate between the different levels of collision danger in each frame with high levels of precision.
- Training time - in the field of collision detection, the faster the model makes a decision, the more time there will be to perform the necessary manoeuvre. So, faster training time indicates faster operation while in use.

1.5 Structure

The structure of this thesis is organised into the following chapters:

- Chapter 1: Introduction to the main topics of the project and formulating the research questions. In this chapter, the research gap and motivations for the thesis will be examined, too.
- Chapter 2: Examination of relevant literature in the field and highlighting key insights. Here, the problems of physically accurate optical flow and collision detection will be examined through the lens of previous literature.

- Chapter 3: Explaining the design and structure of the proposed research method and experiments. This will include two models for predicting the motion of an object in the image plane and several for the collision detection system.
- Chapter 4: Presenting the results of the experiment, where accuracy and time to train the models will be displayed, and one example of complex motion. Simultaneously, a detailed examination and interpretation of the results is conducted. Furthermore, insights into the effectiveness of the system will be defined from the results of the conducted experiments.
- Chapter 5: Conclusion of the findings of the thesis. This chapter will summarise the findings and discuss the limitations of the experiments and possible future work.

2 Literature Review

This chapter aims to present the relevant academic research on this thesis topic. It starts with a justification of the idea that optical flow can carry physical properties in section 2.1. Then the main architecture of this thesis is explored: PIML and specifically the Hamiltonian neural network in section 2.2. Following that, different approaches to collision detection are examined to determine the structure of this thesis's system for collision detection in section 2.3. And finally, the main dataset and relevant previous work on it will be presented in section 2.4.

2.1 Physical Optical Flow

This section will discuss several domains where the optical flow was used in physics-based applications to show that optical flow is capable of representing physical qualities.

2.1.1 Kalman Filter

In every aspect of modern life, technology uses sensors to gather information about the real world. These sensors use many different approaches to estimate some quantity of the real world. Unfortunately, as precisely as these sensors interpret the information, there can be outside factors that are not measured by the sensor that can skew their perception, which leads to wrong measurements. With so many different sensors, there is a need for a generic approach that numerically improves the quality of the perceived measurements by approximating the unobserved parameters. The solution came from Rudolf E. Kalman in the form of the Kalman filter [29]. The best use of the Kalman filter consists of combining noisy measurements with a predetermined model of how the measurements were expected. This combination is necessary because the truth usually lies in the middle. The sensor can produce a noisy measurement, and the model cannot account for all factors affecting the system. The Kalman filter describes the motion with the state transition equation:

$$\mathbf{x}_k = \mathbf{F}_k \mathbf{x}_{k-1} + \mathbf{B}_k \mathbf{u}_k + \mathbf{w}_k \quad (1)$$

Where:

- $\mathbf{x}_k$ is the state of the system at time k
- $\mathbf{F}_k$ is the state transition matrix which represents the model of the evolution of the system.
- $\mathbf{B}_k$ is the control input matrix that modifies the sensory input for the specific system.
- $\mathbf{u}_k$ is the control vector or the sensory input
- $\mathbf{w}_k$ is the process noise that encapsulates all unmodeled disturbances that impact the evolution of the system.

The Kalman filter algorithm consists of several key steps, which can be seen in (Figure 1)

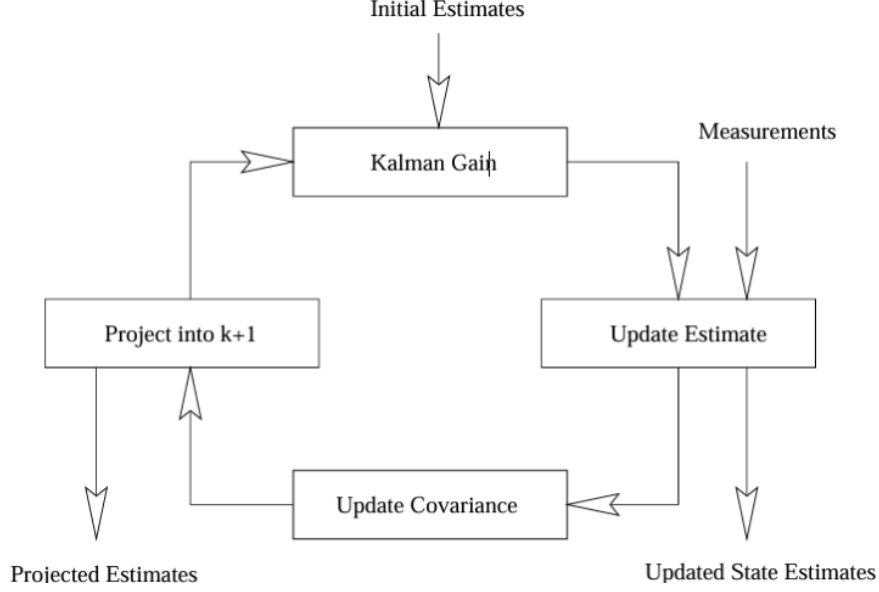


Figure 1: Kalman filter algorithm. Here, the main steps of updating the estimate and the covariance, state transition and Kalman gain estimation are illustrated. Image taken from [34]

Kalman filter has several different components to describe the evolution of the system. First is the state vector that describes the object of interest in one time step. Next is the state transition model that describes how the measurements should evolve in time. For the sensor measurements, there is a measurement vector. And finally, there is the covariance matrix that measures the uncertainty and the correlation between errors. The Kalman filter algorithm is characterised by five steps, and will be looked at from the lens of the problem in this thesis. The first step is to initialise the system state by getting the first measurement. The system state error covariance will be set to the accuracy of the first measurement.

The second step is to reinitialise the system state. This is to take one more measurement. Here, the system state error covariance will be set to the accuracy of the second measurement and the accuracy of any measurement that can be described with the difference between the two states, which is, in this case, velocity.

In the third step, also called the prediction step, by adding the third measurement, the transition matrix is calculated. This matrix is used as a constant velocity model in this thesis's example, which can determine the expected state at time k . State Prediction:

$$\hat{x}_{k|k-1} = F_k \hat{x}_{k-1|k-1} + B_k u_k \quad (2)$$

Here $\hat{x}_{k|k-1}$ represents the predicted state at time k and $\hat{x}_{k-1|k-1}$ is the updated state. F is the state transition matrix while B is the control input matrix of the input $\mathbf{u}_k$. Covariance prediction:

$$P_{k|k-1} = F_k P_{k-1|k-1} F_k^\top + Q_k \quad (3)$$

Here $P_{k|k-1}$ is the predicted error covariance while $P_{k-1|k-1}$ is the updated one. Also $\mathbf{Q}$ is the process noise covariance matrix. This matrix represents the uncertainty of a state that is caused by unobserved forces that are not modelled by the system model.

The fourth step, also called the update step, is to calculate the Kalman gain, which is how much uncertainty a specific measurement adds to the system state estimate.

$$K_k = P_{k|k-1} H_k^\top (H_k P_{k|k-1} H_k^\top + R_k)^{-1} \quad (4)$$

Here $\mathbf{K}_k$ represents the Kalman gain at timestep k . Also, $\mathbf{H}$ is the measurement model matrix and $\mathbf{R}$ is the measurement noise covariance matrix. This means that noisy measurements will not gain much trust because, in their situation, the current state will be considered more accurate.

In the fifth step, the Kalman gain is used to calculate the current state and error covariance matrix at the time of this measurement.

$$\hat{x}_{k|k} = \hat{x}_{k|k-1} + K_k (z_k - H_k \hat{x}_{k|k-1}) \quad (5)$$

Here $\hat{x}_{k|k}$ is the updated state, while $\hat{x}_{k|k-1}$ is the prior state and $\mathbf{z}_k$ is the measurement at time k . Here, the Kalman gain is used to scale the measurement and compute the current state and error covariance matrix. The error covariance matrix is updated to represent the change in uncertainty after incorporating the measurement.

$$P_{k|k} = (I - K_k H_k) P_{k|k-1} \quad (6)$$

Here I is an identity matrix. From this, the algorithm continues to step three for every new measurement.

The relevance of the method to this thesis is that the model for the evolution of the behaviour is usually represented by a physical model of the motion in physical settings. In that case, Kalman filter can be used on optical flow predictions to stabilise them to be more physically accurate.

The field of object tracking is perfect for incorporating physical priors in computer vision. The movement of an object of interest follows a trajectory that, in many cases, can be described as a model of the motion. The method proves to be useful when the object becomes obscured and sensory data is not available. One such case is the paper [8], where the combined optical flow from several frames is used as a model in a Kalman filter. This model provides object motion estimation, which the Kalman filter uses to predict the location of the tracked object in the future. A different example is a paper [49] in which an optical flow propagation model is used to improve the accuracy of optical flow measurements. Furthermore, [4] increases the accuracy of the estimated optical flow with an accelerated motion model for the Kalman filter. In both examples, some kind of consistent motion model was needed so that the Kalman filter could improve optical flow measurements. An even more interesting example is [30] where optical flow is used as sensor data, and more importantly, Newtonian laws of motion are used as a state transition model. In this example, the models of motion are not just implied to be physically accurate, but they are governed by Newtonian mechanics, which defines humans' understanding of motion in the world. These ideas are proven to be useful for collision detection, because it is possible that the problematic object disappears out of the picture for several frames, but it still needs to be considered.

With all this in mind, it is clear that optical flow estimation can work in conjunction with physical modelling techniques of motion, like the Kalman filter.

2.1.2 Temporal Coherent Optical Flow

Motion in the real world can be described with the constant shift between kinetic and potential energy. This shift is caused by the different measurable forces acting on the object. In this case, the change in motion will follow a predictable model of behaviour dictated by the forces acting on it. That said, in theory, in a mechanical system with knowledge about all the forces acting on an object, the motion can be described with a single model. Such consistency in the change of motion is another field where optical flow can be modelled in a physically accurate way. If optical flow measurements can be transformed into temporally coherent motion, they can also represent movements from the real world.

The most defining property of temporally coherent flow is generally smooth changes in the data. This is a sign of sensor measurements that are closely related to the ones in nearby frames. In [4], temporal cohesion was used in order to generate more robust and accurate optical flow. This shows how more coherent in time optical flow is a desirable state to better represent the real world. Therefore, optical flow is estimated as invariant state vectors. These vectors are able to predict the change in motion in future frames based on the model of the motion. In the paper, the authors proposed explanations for the lack of temporal coherent optical flow methods because of the difficulties of balancing smoothness terms for the optical flow and the temporal cohesion. In [31], “inertial estimates” are added to estimate the flow based on the neighbouring frames. These estimates follow the principles of the physical inertia, which is able to smooth large deviations between frames. This makes the optical flow behave like having real physical inertia. With all that said, smooth optical flow can better represent motion that is coherent in time.

Optical flow often suffers from noisy measurements at different time steps. These incorrect batches of data are able to distort the otherwise consistent change of motion. One way to impose temporal coherence is to add regularisation terms and trajectory smoothing terms to ensure that this motion will be stable and explainable between frames, even in the case of noisy measurements. Many of these regularisation terms have physics backing, like the Euler-Lagrange equations used in [5]. In [56], the authors proposed a method that uses trajectory regularisation on optical flow extracted from multiple frames. The authors claim that flow in a sequence obeys a certain trajectory motion model, so if the motion is regularised with this model, it can produce more stable results. This displays that regularised optical flow can be described with a single trajectorial model, which is the behaviour of a physical trajectory.

All the above developments show that several consecutive optical flow measurements can be described with a physical equation.

2.1.3 Fluid Dynamics

The motion of fluids can be described with many different particles that simultaneously move according to some physical equation of motion. Another field where physics-imbued optical flow is used is in modelling fluid dynamics. This means that if fluid is modelled from optical flow measurements, these measurements follow fluid dynamics physical models. Two of the most common methods to model fluid dynamics are particle image velocimetry (PIV) and particle-tracking velocimetry (PTV), which can determine the

movement of the flow from designated particles that are observable.

In [57], the authors mentioned that the brightness constraint on which optical flow is constructed has no physical backing, but despite that, the equations for the motion of optical flow can be transformed into a physical equation. This can be seen in the representation of the physical vortex motion using velocity vectors from optical flow (Figure 2). The evidence for this was given by the observation that optical flow was proportional to the path-average velocity of fluid or particle in flow visualisations.

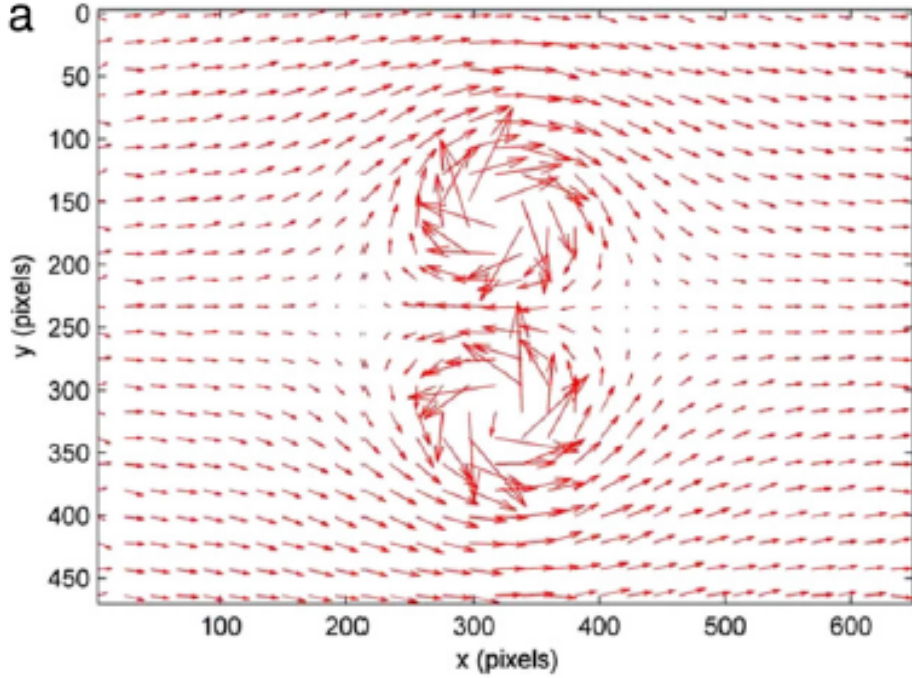


Figure 2: Velocity vectors extracted from particle images. Image taken from [57]

Furthermore, in [36], the author establishes a quantitative connection between the optical flow and fluid flow, showing that measurements from optical flow can be used for scientific measurements of physical events. The author stated that optical flow does not have a clearly defined physical meaning, so it can not be used directly in engineering problems. Despite that, they also expressed positivity that optical flow can be used for high-resolution velocity fields because physical and geometric constraints can easily be incorporated into optical flow solutions.

One great explanation of optical flow in the real world is given in [11]. Here, the author explains that image data only provides physical quantities indirectly, and the computer vision approaches are used to extract them from the image. To do so, a regularisation function is used to preserve divergence and vorticity blobs of the motion field.

With all this in mind, it is clear that optical flow with additional transformation can plausibly model fluid dynamics. And because of this, optical flow represents the fluid dynamics of the system.

2.1.4 Summary

To summarise, even though the 3D motion of the world is transformed to a 2D image representation, the data captured by this transformation is still based on physical principles. Methods like Kalman filtering or fluid flow estimations that are meant for sensory physical data are applicable to modified optical flow data. This gives a strong argument for the premise proposed in the thesis that optical flow, despite being only an **approximation of physical motion**, can be modelled as one.

2.2 Physics Informed Machine Learning

In this section, commonly used PIML approaches will be examined to determine which ones to use in the work of this thesis. Based on the paper [3] that considers the strengths of PIML in the domain of computer vision, PIML has many advantages over standard computer vision approaches. First, PIML makes the models more mathematically consistent because it is based on the mathematics of the processes in the real world. At the same time, it can generalise only from a couple of points of data, so it uses dramatically less data to operate than standard computer vision approaches. Furthermore, with clearly defined mathematical equations, there are fewer variables to optimise. In that case, with reduced dimensionality of the search space, convergence is reached much faster. Yet the greatest advantage is that these models are much more generalizable and less data-specific since they use underlying physical principles rather than commonly seen patterns. And of course, if the underlying physical equation of a problem is known, the models are more interpretable by human engineers.

In [48], the two opposing approaches of physics-based modelling and neural networks were compared. Physics-based modelling, on the one hand, is quantifiable, measurable and needs less data. On the other hand, neural network solutions can better adapt to changing circumstances or unknown situations and are computationally faster. As with many problems, the best approach is to combine both fields' strengths. The most widely used method that combines physical models and neural networks is to incorporate physical constraints into the loss function of the NN model. This way, the model becomes biased to physically correct solutions. This is done so the model is trying to minimise two things in the loss function at the same time. One is to have the prediction as similar as possible to the data, and the other is to have a prediction that satisfies well-known physical equations. This is the general principle in physics-informed neural networks (PINN) [46], (Figure 3). With all this said, the approach that models some physical properties with a modified loss function ought to be the best choice for this particular thesis problem.

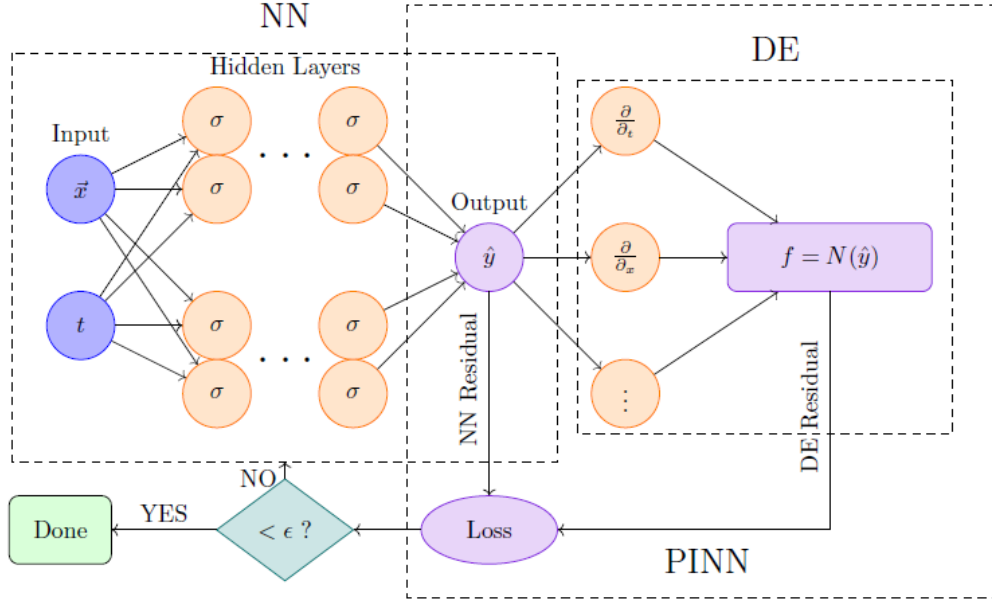


Figure 3: General example of PINN architecture. Here, NN represents the standard neural network with a loss function only on the output. On the other hand, DE represents the additional loss function that adds to the final loss of the PINN.

To better separate different methods based on the level of enforcement of physical laws, the work in [19] differentiates physics-guided, physics-informed, and physics-encoded neural networks and operators. The more the model relies on data, the more its outputs rely on statistical models and not concrete and interpretable physical ones, but at the same time, it finds it difficult to model different discrepancies. In this thesis, in particular, it is not expected that be many discrepancies since incoming obstacles in the air usually move in close to constant trajectories. In that case, a model more reliant on physical models will be preferred. At the same time, this experiment will have to work on optical flow data with low physical backing, so rigid governing equations and boundary conditions can impact the performance. Thus, for this thesis, a physics-informed approach will be preferred.

In [18], more approaches to embedding physical laws in neural networks are shown. Feature selection is a method of selecting features that can hold physical meaning from domain knowledge. This can be, for example, temperature, pressure, and velocity in fields like fluid mechanics. Because RGB values of a video do not inherently hold physical information, some kind of features need to be extracted from it. At the same time, the authors emphasise designing features for temporal coherence when they are applicable. To take this into account, developing a representation of the flow from a set of several frames will be necessary to represent the time evolution of the motion.

Another interesting idea proposed in [18] is where physical information can be added to the training data before the training process, with the addition of mathematical symmetries. Unfortunately, object trajectories do not have explicit symmetries that can be used. A different approach is model engineering, which looks at the architecture of the model to add physical information. An example is how recurrent neural networks are

used to model temporal information. An additional technique to limit the solution space only to useful domains is to incorporate regularisation terms. For this thesis, the motion described is quite simple, so there will not be a need for a novel architecture or added regularisation.

Based on looking at many possible ways to use PIML in this experiment, some key insights were collected. First, feature selection needs to be used to find appropriate features to represent the motion and its evolution in time. Second, a model with a physics-informed loss function has to be implemented.

2.2.1 Hamiltonian Neural Network

A cornerstone question of our understanding of the natural world is how to know if any theorem or observation is true if all of them are interpretations of real-world phenomena. A sure way to put a structure in these interpretations is to add a set of invariant quantities or some fundamental principles that will be true in any situation. A commonly referred invariant quantity is the total energy of a system. It states that energy is not lost, but it can only be transformed from kinetic to potential. As for the famous Noether’s theorem [32], which states that any kind of conserved quantities in nature give rise to symmetries and vice versa, this conservation of energy can be observed in the real world. One such example is the orbit of planets that are always symmetrical and do not change without an external force.

The theory that best encapsulates the idea of conserved quantities and motion is Hamiltonian mechanics. The improvement that Sir William Rowan Hamilton added to the established Lagrangian mechanics was that the Hamiltonian function encapsulates both the potential and kinetic energy. It also improved on the Newtonian mechanics that relied solely on forces to describe the world, which were, in many cases, quite difficult to estimate.

The main physical principle that governs the trajectories of incoming objects is the conservation of energy. Because of that, the choice of architecture needs to be the main concern. With all this in mind, the specific PIML architecture for this thesis will be Hamiltonian neural networks [24] (Figure 4). This particular model uses the Hamiltonian equations of motion to train a neural network to approximate energy-conserved motion from observational data.

$$\frac{d\mathbf{q}}{dt} = \frac{\partial H}{\partial \mathbf{p}}, \quad \frac{d\mathbf{p}}{dt} = -\frac{\partial H}{\partial \mathbf{q}}, \quad (7)$$

This system of two partial differential equations with unknowns $\mathbf{p}$ and $\mathbf{q}$ is Hamilton’s equations of motion, where t represents a particular time step. $\mathbf{q}$ represents generalised coordinates of the system or its position in its configuration space. $\mathbf{p}$ is the conjugate momentum or the motion of the system. Combined, they can describe any physical system. The philosophy is that the coordinates describe the current state, and the momentum describes the change in that state to the next one. With these two measurements, it is estimated that the main benefit of this type of mechanics is the Hamiltonian function H . This function represents the total energy of the system as a whole. This function has the property that, by taking its partial derivatives, the time derivatives of the system can be estimated. These derivatives represent the evolution of the coordinates and the

momentum in time. For example, if a ball is thrown in the air vertically, the generalised coordinates will be represented by the height of the ball. This is where the ball is in each time step. The time derivative with respect to the generalised coordinates will be the velocity of the ball or the change of height. The conjugate momentum $\mathbf{p}$ will be the momentum of the ball or velocity times mass. And finally, the time derivative with respect to the conjugate momentum $\mathbf{p}$ will be all forces acting on the ball, like gravity or air resistance. From the paper [24], two major problems of traditional neural network approaches are solved by this architecture: First, the change of motion is continuous, so the best way to express the motion of a specific time step is to model the motion as a set of differential equations. This is different from how machine learning approximates because it does not model the motion as one specific equation. The second problem is that small mistakes in the modelling accumulate over time and give drastically incorrect results. This is because in systems that evolve over time, each new prediction is based on the previous one. In that case, inaccuracy in one prediction is carried over to each subsequent one, so even one percent inaccuracy in every prediction accumulates drastically over time. This inaccurate prediction represents a much different amount of energy in the system than the real one. The conservation of the energy-like structure of the Hamiltonian model is much more resilient to this kind of discrepancy than standard ODE models. This is because the energy can only be converted from kinetic to potential and vice versa, but no new energy can be added, so this kind of accumulating new energy will be impossible.

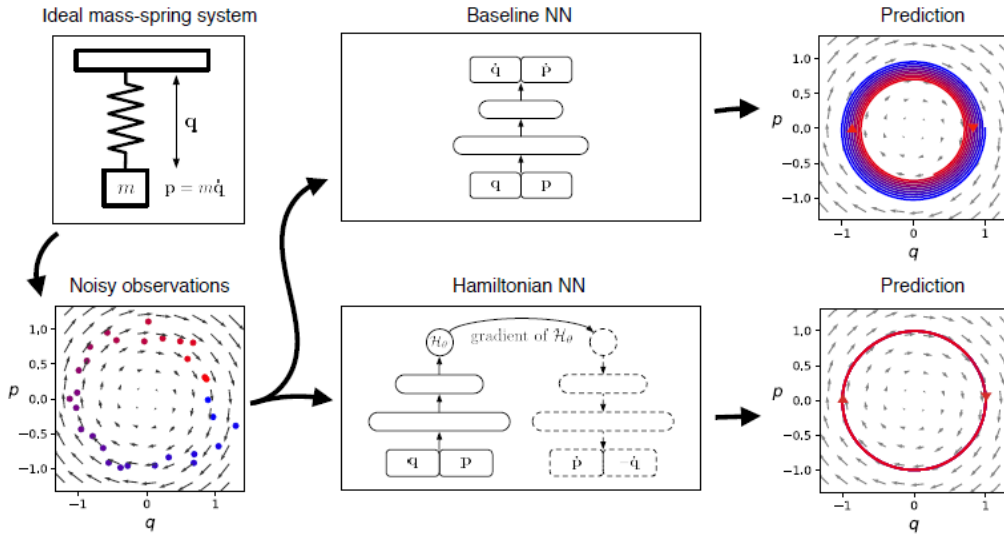


Figure 4: This is a common example for the Hamiltonian neural network. Here, the baseline NN predict the values for $\dot{\mathbf{p}}_i$ and $\dot{\mathbf{q}}_i$ directly from data. This leads to scattered prediction. On the other hand, the Hamiltonian NN estimates the Hamiltonian of the system from which $\dot{\mathbf{p}}_i$ and $\dot{\mathbf{q}}_i$ are derived. This leads to a prediction that is stable and better represents the underlying motion. Image taken from [25].

A Hamiltonian neural network is meant to handle measurements of position and velocity, so the raw video data can not be used directly. In [20], one of the most important problems in PIML is touched upon: how the choice of the coordinate system improves the training of PIML models. With the Hamiltonian neural network architecture, the

latent space is four specific measurements, the generalised coordinates $\mathbf{q}$ and conjugate momentum $\mathbf{p}$ and their time derivatives. These parameters are quite simple in nature and can be derived from any kind of data describing a motion

In this thesis, the motion needed to be modelled is an aerial or ground trajectory of an object in the real world. In [38], a different approach is used where, instead of learning from data, the model finds possible trajectories that can satisfy Hamiltonian equations. This is closely related to the problem of this thesis, which is that the trajectory of an object needs to be derived from a Hamiltonian neural network. Furthermore, [21] shows one big strength of the Hamiltonian neural network model, that is, the resilience to vanishing and exploding gradients by the innate structure of the Hamiltonian equation for conservation of energy. This will allow for the consistent prediction of the trajectory even for several time steps in the future.

From all the evidence provided, a Hamiltonian neural network will be capable of modelling the motion of objects from several consecutive frames while conserving the energy of the system.

2.2.2 Physics Informed Computer Vision

The next area needed for this thesis project is to combine the fields of physics-informed machine learning and computer vision. Therefore, related work that combines these two fields will be explained. Computer vision is able to represent information in the real world in a format that is easy to interpret by machines using a camera as a sensor. Despite other sensors like LIDAR being more accurate about the obstacles in the real world, they are usually too heavy for many UAVs. This makes computer vision applications with **monocular camera** more usable for smaller-scale or more available UAVs. This is why for this thesis, the system will be designed to work on data only from **monocular camera**. If there is a reliance only on a monocular camera, there is a need for clues that can predict a collision.

In the case of collision detection, useful clues will be: **trajectory, size and position on the image plane**. **Trajectory** is the key to describing the movement that considers most of the dynamic obstacles. **Size** can describe the expansion of the object on the screen, so it will give clues about the time to contact. Finally, the **position of the object** can tell if the object is heading to the middle of the screen, which, in general, will guarantee a collision. This is because if an object is in the middle of the screen and moving closer with constant motion, this represents a trajectory directly through the camera, which means collision. With all this in mind, the **position and size of an object on the screen** are a valuable part of the necessary layout space for this thesis's system.

The first area that needs consideration is learning physical properties from a video. (Figure 5) In fields like computer vision, where the input can be an enormous amount of numbers representing RGB values of the pixels, there is a need to find a much simpler representation of the data. To tackle the problem, aggregation methods are used to reduce the data to components that can represent only specific features of interest. This is called latent representation, and based on the choice of representation, different insights can be extracted from the data. In the papers [9], [13], [58], the main architecture con-



Figure 5: In the first image, there is a simulation of two different objects moving through each other and colliding. The second image is the estimated trajectory from a machine learning model in each time step for the two objects learned only from simulation video data. The other insight gained in the second picture, which is much more relevant to this thesis, is whether this trajectory causes a collision. Image taken from [13].

sists of some type of latent space representation extraction of the movement of objects from several frames. From this representation, a prediction or simulation can calculate future motion or outcomes. Incorporation of the idea of the latent space of the physical properties is crucial since it can represent the needed information to model a specific motion.

From the vast amount of data humanity has collected, only a very small amount has **clear annotations** about what is represented. This provides the need for techniques for using unprocessed data. A similar idea but much more relevant to this project is the paper of [15], which, instead of labelled data, uses several unsupervised techniques to extract the latent space of the physical properties from a video. This way, it can be used on a wider range of datasets and can provide more generalizable solutions not confined to the simulation ones. A similar way for unsupervised gathering latent space is [59], which represents the real-world perception module and then recreates it with physics and graphics engines. An even better example of extracting latent space from the video is [26], where, in addition to the latent space of the physical properties, the model also learns one additional latent space. This second latent space divides the information from the image into motion and the semantic meaning of the image. In a different area of study but quite relevant to this thesis is this paper on video compression [39]. In it, the proposed method finds the physically accurate motion of each segment in a video for multiple frames, so it can be compressed to only the objects and their motion. Here, the motion follows Newton’s laws of motion and is calculated on a segmented video per object. These motions describe the physically informed motion of an object between several frames. This shows how motion estimated from a video source can be physically accurate without any additional annotations for the data. From all that is mentioned in this paragraph, there is no need to find motion representation in raw data without human annotations.

Nevertheless, PIML has to handle trajectories of real-life objects as well as simulated ones. This is required to be used in computer vision applications in the real world. Some examples include [37] and [22], which show how PIML can model trajectories of ground

and aerial vehicles from information gathered from a video camera. The dataset used, "NGSIM", consists of trajectories of vehicles on highways constructed from their displacement between frames, which is close to the setup of this experiment. Another example is [10] where the authors model the trajectory of a table tennis ball with PINN using information from a single camera. All these examples show already proven methods of modelling trajectories in a physically accurate way.

One very promising example, but unfortunately using a depth camera, not a monocular one, is [60], where a system for autonomous vision-based navigation is presented with prior knowledge of the physical system. If not for the different kinds of sensors of choice, this work enables the incorporation of physical knowledge into a video understanding system to prevent collisions with moving and static obstacles.

Overall, PIML has already proven its effectiveness in real-life computer vision applications, with many more to come.

2.3 Collision Detection

After the motion from optical flow is successfully modelled, these measurements can be used to make predictions in the real world. One field where these predictions can be examined is that of **collision detection**. This considers techniques that can show if an object in the field of view is moving through the fixed position of the camera. There have been many research ideas for collision detection, with some of them using optical flow and some not. Some models that do not use optical flow, like [28], use very domain-specific data and are not generalizable. Others like [1] use a method of expansion of feature points. The idea here is that first, feature points are calculated at points of interest, and then the expansion rate of these points can indicate that the size of the object in the image plane is increasing. This size increase can indicate that the observed object is getting closer to the camera. This approach does not consider the trajectory of the object, which means the alert will come at a stage where the object is close enough that its change in size is noticeable. Some successful ones, like [52], use several whole frames in LSTM architectures. This is quite computationally heavy when the key points of the movement are few in the image. And above all, these methods do not actually base their assumptions on the physical movement of obstacles, but more on visible changes when an obstacle is becoming closer to the camera.

The first method used for collision detection using optical flow is **time to contact** [45], [42], [54]. This method calculates the remaining frames before collision with an object by examining the enlargement of the object in the captured video. With optical flow, "focus of expansion" can be observed, which is a point at which vectors point outward, showing that the object is expanding in the image. Another important factor is the ratio of expansion because closer objects have a higher ratio than faraway ones. Finally, the position of the focus of expansion and the magnitude can predict whether the object will collide or just pass by. If the position of the focus of expansion is closer to the centre of the image, this can indicate a collision. From this, it can be derived that expanding objects near the centre of the image can indicate a possible collision. This can be seen in the example with three different trajectories of simulated planes in (Figure-6)

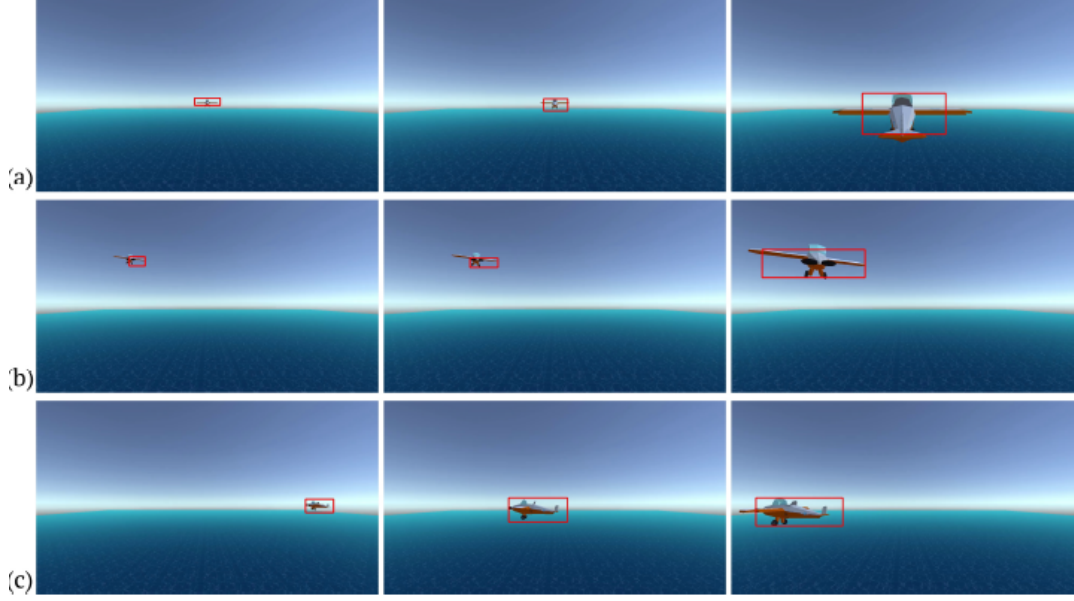


Figure 6: Three different examples of dynamic obstacles with different trajectories and positions on the image plane. An important one is example (a), which shows how a centred object on the image plane passes by the position of the camera in an experiment, which indicates a collision. Image taken from [55]

The second most widely used method is **motion clustering** [12], [50]. The idea is that pixels from one object will move in the same direction, so a clustering based on the motion can give us insight into the direction that the object is going. If there are batches of the optical flow of an object, different moving directions can be deduced, not just left or right, but through the UAV and away from it. This is deduced by the expansion of the flow in the cluster because the closer the object is, the more intense the flow is. One advantage here is that by using optical flow, the motion of the camera (called ego motion) can be excluded and focus only on the motion of the object in the image. These 2D optical flow clusters can also provide other 3D motion information on the object, like translation and rotation, that can be calculated by several points in a cluster. After the objects are clustered, a potential obstacle is a region where the magnitude of movement is higher than the global average. This can signal a moving object or movement toward an object. The big advantage of clustering the image is that it drastically reduces the input data. This is due to the fact that the motion of a cluster is generally similar, so there is no need for all optical flow measurements to describe it. Even better data from the clusters is perfect to incorporate into the previously mentioned Kalman filters [33] for better tracking of the segments. This is useful because it shows that the motion of an object is usually constant through the whole region of the image plane occupied by the object. Consequently, the object's motion can be examined only at a few points. This can drastically reduce the amount of data that needs to be modelled to have a full perspective of the general motion of an obstacle.

From all the research in the area, it seems that there are two important components for collision detection. First is clustering the motion to isolate the motion of a particular object. This can be done using video segmentation for different objects in the image plane. The second one is searching for expanding optical flow and flow consistently directed to

the centre of the image plane of the camera.

2.4 ColANet Dataset

The dataset in use for this thesis is ColANet [44], which contains one hundred videos of UAVs crashing into moving and static objects both in the sky and on land. The dataset offers a wide variety of weather conditions, from sunny to cloudy. All the frames in the dataset are labelled as either free fly or collision, with a total of over 2000 collision frames and 6000 free flying frames. Another aspect of it that was perfect for this thesis is the mixture of static and dynamic obstacles with different sizes and speeds.

To determine the best architectures to compare the proposed one in this thesis, previous works on this dataset need to be considered. Due to the nature of the dataset, the conventional CNN model will be preferred because they are often used to extract structure from visual data. In [62] and [61], several pre-trained CNN models and the possibility of using transfer learning between models to improve performance are discussed. In [14], object detection with YOLO was used in conjunction with the R-CNN model to reduce the computational resources. In [43], the architecture used was a hybrid neural network that uses CNN for extracting visual features, a recurrent neural network(RNN)to process temporal dependencies and a feedforward network for determining collision probability.

From all the previous work on this dataset, comparable models need to be able to extract meaning from the visual data with the CNN model and need to be able to handle temporal dependencies with RNN.

3 Methodology

Design choices of a study shape the direction of the entire research and the outcomes it tries to achieve. From the choice of data and its sampling to the type of conclusions that are sought, this all contributes to a cohesive and full argument(Figure 7). This chapter first looks at the general design choices of the thesis in section 3.1. Following this, the whole process of data preparation is described in section 3.2. Finally, the methods of modelling the data is presented in section 3.3.

3.1 Research design

The real world is governed by physical laws, and the methods that model them have to be based on empirical evidence and measurable data. In that case, the thesis focuses on the observable phenomena of optical flow and minimises any subjective interpretations of the data. Therefore, the research philosophy of this thesis follows **objective positivism**, which represents the first layer of the research onion presented in (Figure 7)

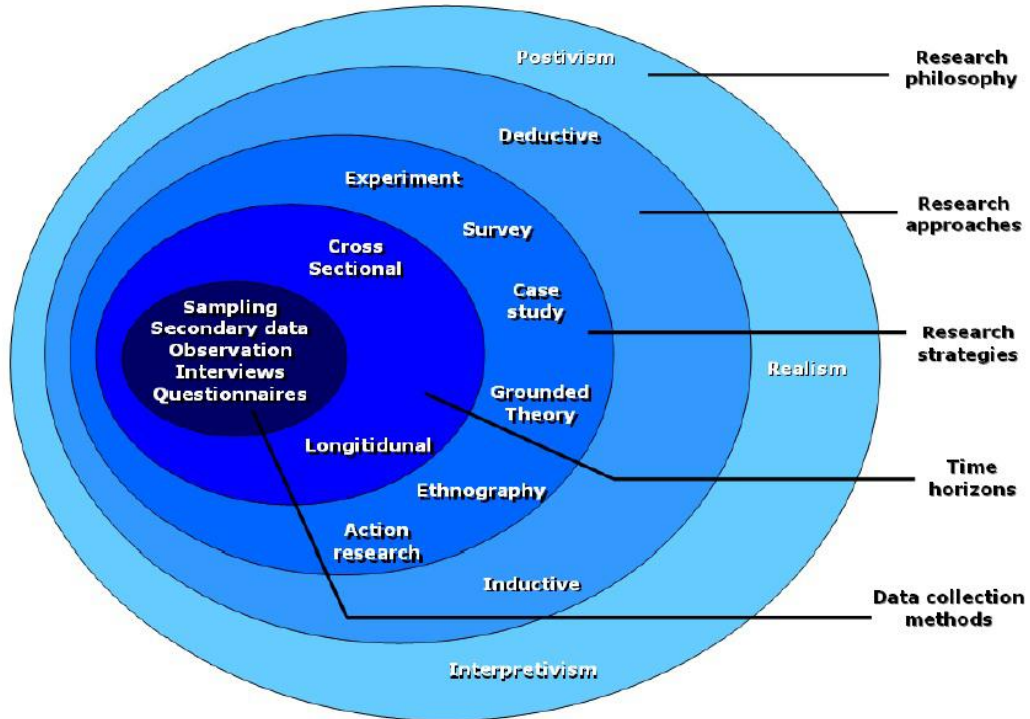


Figure 7: The research onion explaining possible research design choices. Image taken from [53].

The design for the system of this thesis came from the theories of physics-informed machine learning, with experimentation of creating a new approach. Going from theory to experiments, this makes the research approach **deductive**.

The nature of this research is **empirical and logical**. The measurements of motion gathered by optical flow are empirical data that is based on fundamental principles of how changes in position can predict movement. Unfortunately, due to the lack of research

in the field, there was no well-known way to model or prepare the data, so, many logical hypotheses and design choices were considered that are not present in any literature but follow logical propositions as a basis.

Due to this thesis approach, based on numerical visual data, the most appropriate choice of research design was **quantitative**. The data is based on discrete observations of the state in the image plane in various time steps that give concrete quantities for each state.

The strategy of this research mainly follows the use of **experiments and ground truths**. In the development of a new way of modelling specific data, a lot of approaches needed to be developed and controlled experiments was the way to validate them from the collected data. At the same time, some fundamental truths of any kind of similar methods were found that can be used as a ground truth for future research.

The time horizon of this thesis is a **cross-study**. All of the data was collected at the beginning of it, and no new data was introduced at any stage of the development.

The data was gathered by **convenience sampling**, which is a methodology where appropriate and easy-to-acquire data for the specific domain is gathered. From many other sources of similar data on the internet, this one had a wide variety of terrains and dangers, both static and dynamic. This makes the data quite representative of the field of UAV collision detection. There were no AI generated videos, so it accurately represents the real world. In most cases, the object of collision was clearly visible, so it was possible to focus on it.

The general approach of the study is **exploratory**, which focuses on finding new solutions from theories. In this thesis, new ways to model optical flow data were found. At the same time, all of the processes and result analysis are based on quantitative physical data.

3.2 Data preparation

Data, in general, is the measurement of a sensor in the real world. But besides that, the raw measurement of the sensor can not be used for every single application. The idea of data preprocessing is to filter data in a way that holds some wanted truth in it, and transforms other parts that are less useful. In machine learning, a lot of the time, there is no need for all the information in the data, but some compressed latent space that can be used to model specific behaviour. In the context of physics-informed machine learning, pixel values can not show any kind of physical quantity by themselves, but the comparison of groups of points of interest between time steps can describe a physical phenomenon. In the case of this thesis, there is no explicit measurement in each frame of how much danger the UAV is in. For this reason, a **custom scoring function** was developed to label each frame. The pipeline of different consecutive steps to preprocess the data is shown in (Figure 8).

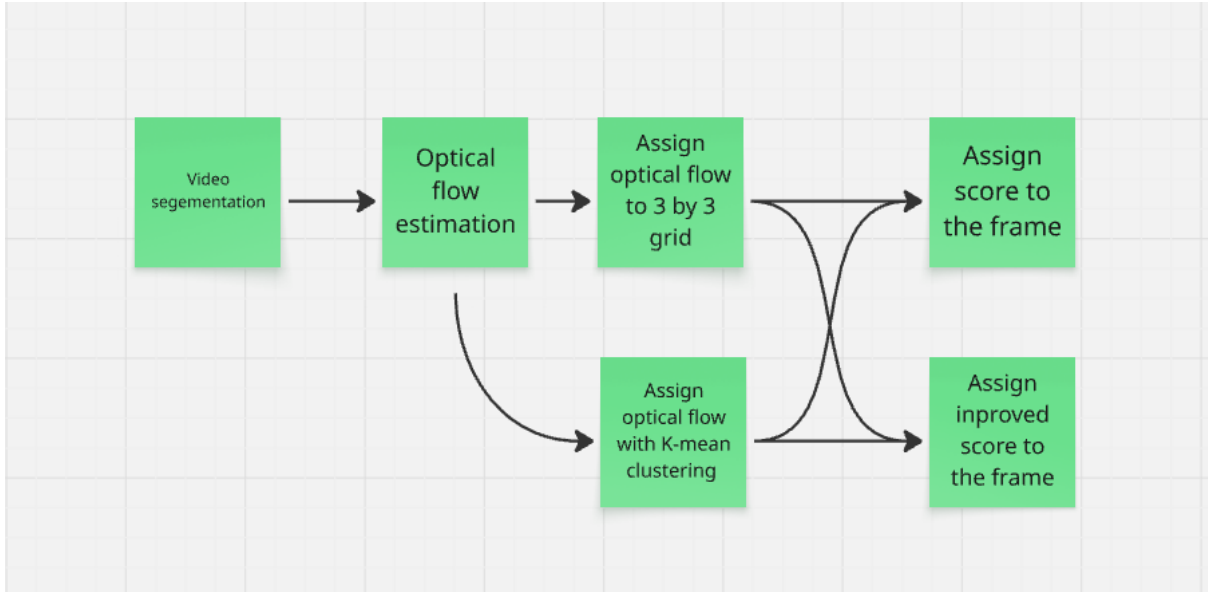


Figure 8: Pipeline of the data preprocessing methods of this thesis.

Video segmentation is explained in section 3.2.1. After that, optical flow estimation and assignment to the three-by-three grid will be examined in section 3.2.2. Furthermore, the assignment with K-means clustering is described in section 3.2.3. Finally, the scoring functions are shown in section 3.2.4.

The dataset used for this project is ColANet [46] described in section 2.4. The dataset was chosen because of the diverse range of objects, their speed and conditions in the videos. It also contained some rotation motion of the UAV and videos with several moving objects for more challenging examples.

3.2.1 Segmentation

The first part of the preprocessing of the data was the isolation of different sources of motion. In many videos, more than one of the objects in the field of view is moving. The reliance on points of interest in different objects made segmentation the only way to isolate a particular motion. This way, the only information about one source of change is present at a time and can be used to extract the latent space from it. To explain how segmentation works in general, after selecting an area, a mask with the distribution of colours in that area is made. Then this mask is applied to every consecutive frame in different places, and in the region of it with the most similar distribution to the one of the mask is selected as the next iteration of the selected object. To achieve this, the Meta’s SAM2 [47] video segmentation model was chosen. An example result is examined in (Figure9)



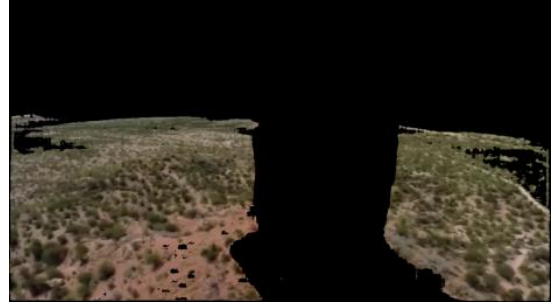
(a) Original image



(b) Segment 1



(c) Segment 2



(d) Segment 3

Figure 9: Example of an image segmented with SAM2 in three main regions in the image plane. In the segmented image, the colored area is the segmented object, while the black area is the excluded part of the image.

This particular model was the best choice because it is an open-source model with great performance in video segmentation benchmarks. Furthermore, it is effective due to its zero-shot generalisation that allows for segmenting arbitrary objects in videos without requiring additional task-specific training data. Another big advantage was the speed of processing a high amount of data. By far, the most important quality that was used in the project was its ability to propagate the mask through frames, even when more challenging ones appear. For the specific configuration, the Hiera(Hierarchical Efficient Representation Learning) backbone was used, because it is the standard and best-performing one. For the choice of model, the large model was used, due to its superior performance on difficult segmentation tasks compared to the medium or small models.

3.2.2 Optical flow

A key concept in physics is the motion of objects in the environment. This is characterised by changes in position in time due to an external force. If measured correctly, the motion can model these changes based on physical laws. In the field of computer vision, the information about the world is condensed to the image plane, so any changes in the 3D world are reflected in a specific way on the 2D image plane. The most widely used method to approximate the movement of a real-life object on the image plane is optical flow. The key assumption this method takes is that brightness does not change between frames. This is because, for example, two darker pixels that are close to each other in consecutive frames have to be the same, but in different positions. This change should be due to motion and not the light becoming lower in different places between frames. There are two types of optical flow: sparse and dense. The main difference is that sparse optical

flow calculates the displacement of pixels in points of interest, which are usually edges and corners, while dense optical flow calculates the displacement of all pixels. With the computational constraints of UAVs in general, the sparse optical flow was chosen because of its lower resource demand.

For the application in the context of this project, sparse optical flow was estimated for every frame of the videos from the training data. Each optical flow measurement is characterised with X and Y coordinates of the point and the VX and VY coordinates of the vector of the optical flow. In the physical sense, the X and Y coordinates of the points of interest were used as the position of the object and the displacement between the frames was taken as the velocity with respect to these positions. Because the mass of the object is not considered, the displacement will be equal to the momentum of that object in the case of this thesis.

The exact implementation of optical flow that was used is `calcOpticalFlowPyrLK` from `cv2` [6]. This implementation is based on the Lucas Kanade optical flow with a pyramid approach. This is a widely used approach that adds an additional assumption that optical flow is the same in a small window, or in other words, that neighbouring pixels move in the same direction. Due to the nature of the problem of collision detection of UAVs, there is no expectation of wider movements, especially in the sky, so this is the perfect choice for the task. One particular strength of this implementation is the pyramid approach, where the image is viewed in different resolutions. In this case, larger movements can be spotted on the low resolution and small refined movements in the higher resolution. This can be seen in figure 10 where it is shown how optical flow is measured in images with different resolutions.

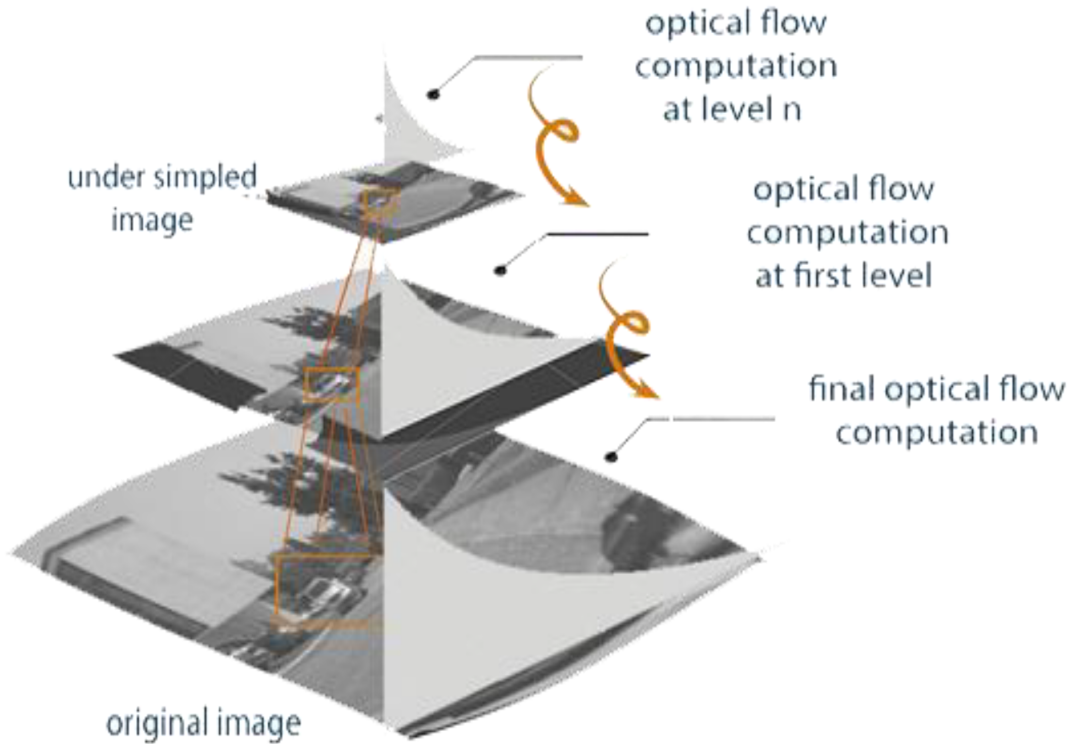


Figure 10: Image pyramid with different image resolutions. Image taken from [2].

The mathematics of this method first considers the vector field (u, v) , which is

the collective displacements of the pixels in the x and y directions. This means that if multiplied by all points in a region, the points will move to a new position based on the magnitude of the optical flow in horizontal and vertical directions. The way that it works is by taking the partial derivatives with respect to the x and y coordinates for every point in a neighbourhood of points P and multiplying them by the optical flow of the neighbourhood (u, v) , the partial derivative with respect to time of these points can be acquired. This is because the brightness is constant from the main assumption, so there shouldn't be any loss of it in time.

$$\begin{bmatrix} \sum_{p \in P} I_x I_x & \sum_{p \in P} I_x I_y \\ \sum_{p \in P} I_y I_x & \sum_{p \in P} I_y I_y \end{bmatrix} \begin{bmatrix} u \\ v \end{bmatrix} = - \begin{bmatrix} \sum_{p \in P} I_x I_t \\ \sum_{p \in P} I_y I_t \end{bmatrix}$$

In this equation I is represented by the function of the brightness of a pixel $I(x, y, t)$ where x and y are coordinates and t is the number of the frame. From these three inputs, the function returns the brightness of the pixel in this location and time step. Next is $I_x = \frac{\partial I}{\partial x}$, which represents the partial derivative in the x-direction. This indicates the magnitude of changes in the horizontal direction. $I_y = \frac{\partial I}{\partial y}$ represents the partial derivative in the y-direction. This indicates the magnitude of changes in the vertical direction. And finally, $I_t = \frac{\partial I}{\partial t}$: which represents the change of brightness intensity between frames.

Optical flow estimation have several hyperparameters that need to be specified. To be able to find optical flow measurements, points of interest need to be chosen. These points are often around edges or corners of objects in the image, where a wide difference in colour can be observed. First, hyperparameters about the corner detection will be examined. The maximum number of corners is set to one hundred, which is enough for this application due to the relatively simple forms of the moving objects. The quality level is set to zero point three with the distance between corners and size of the blocks being seven pixels, which is a standard combination for similar datasets. For the optical flow estimator, the window size is fifteen by fifteen pixels. This was wide enough to catch most movements of pixels while still preventing the collection of noise that can occur with wider windows. The maximum level of the image pyramid was set to two, because again, larger movements are not expected in this dataset. The criterion for stopping the algorithm from iterating over the points is either ten iterations, or zero point three or less difference between estimations.

With the motion being calculated per segment of the image, most of the measurements will likely be similar to one another. In this case, there is no need to consider all of the optical flow measurements to represent the motion.

For this, a new approach was developed inspired by the YOLO segmentation architecture [41]. In this method, the segment is divided into a three-by-three grid, and the average motion is calculated in each grid cell. The necessity to look at more than one point in the image instead of the global average flow is because some specific movements can't be captured this way. For example, a rotation of the object in the image is characterised by motion in all possible directions in every corner of the segment. A more important case for this particular project is "focus of expansion", which is characterised by motion pointing to the middle of the segment. Example of this structure can be observed in (Figure 11).

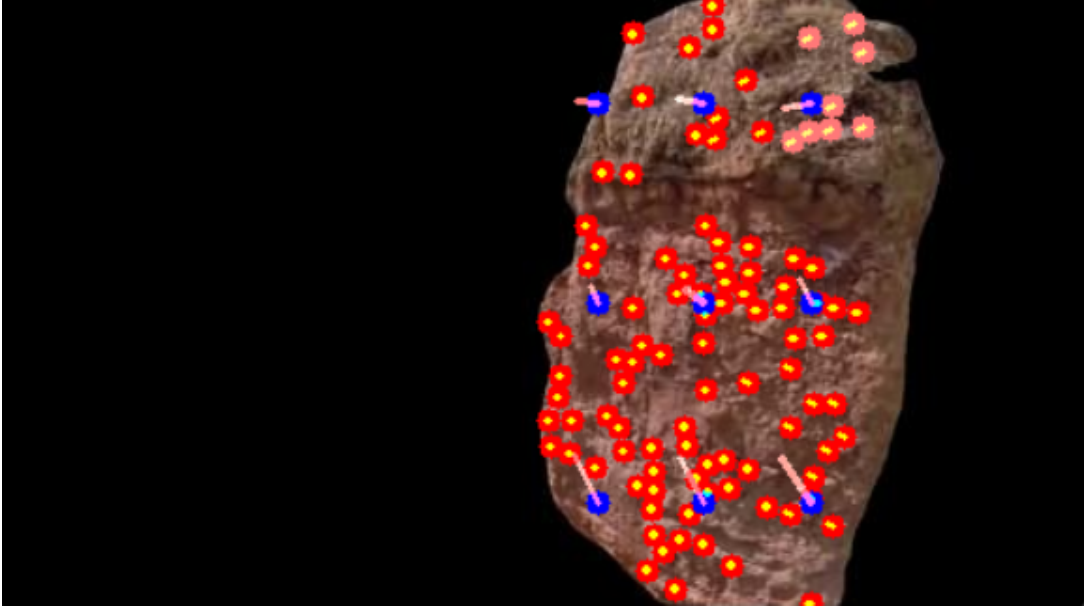


Figure 11: This is an example of the nine-segment grid. Red and pink dots are optical flow measurements. The blue ones are the grid average optical flow from the Hamiltonian prediction about three steps in the future. The pink ones are the optical flow measurements of one of the nine grids. In this image, only the non black area is the observed segment, which in this case is a picture of a rock.

By combining everything mentioned, the standard image characterised with thousands of numbers for the values of each pixel was condensed to a latent space of nine points, each characterised by four numbers: X , Y , VX and VY . This sums up to thirty-six datapoints that encapsulate all of the necessary aspects of the motion of a particular segment.

3.2.3 K means clustering

In nature, motion can hardly be categorised by only the position in space it occupies. For that purpose, one of the improvements over the first approach was adding clustering of the optical flow measurements instead of fitting them based on their position in the image. This approach was inspired by the work in [12]. This way, the flow is combined not only based on where it was collected but also on its direction. This is the case because the clustering algorithm was fed both the position and velocity of each point of interest.

The algorithm of choice for this task was k-means clustering [7]. This algorithm is a simple and fast method to cluster even large amounts of data. Example of this can be seen in figure 12.

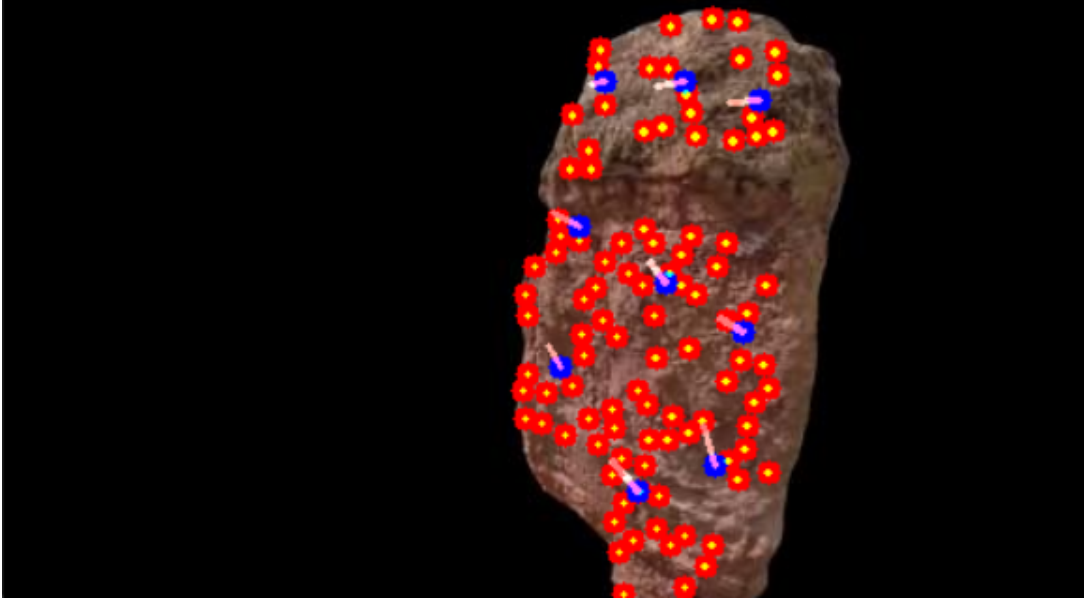


Figure 12: This is an example of the clustered flow method. Red dots are optical flow measurements. The blue ones are the grid average optical flow from the Hamiltonian prediction about three steps in the future. In this image, only the non black area is the observed segment, which in this case is a picture of a rock.

K-means clustering is characterised by five steps. First, the number of clusters is decided, which in this case was proportional to the number of collected optical flow measurements, up to a maximum of nine. After a lot of experiments, the choice for the number of clusters k was decided to be the number of optical flow measurements divided by five, to a maximum of nine clusters. In the second step, all of the measurements are assigned randomly to each cluster. In this case, optical flow measurements are assigned to a cluster. Third, the distance of each observation to each cluster is measured. For optical flow, this is the distance not only in the coordinates but also in the difference of the vectors. In the fourth step, each observation is assigned to the cluster to which this distance is the smallest. And the final fifth step, for each cluster, the average of each of the four characteristics of a point is calculated. Then each cluster is moved to the location of that average result. Steps three to five are repeated for a set number of times to cluster the data more precisely.

K-means clustering was tested for more fundamental reasons, too. In the field of computer vision, segmenting the image will give close enough isolation of a particular motion, but this is not translated into other fields of physics-informed machine learning. Different kinds of sensor observations need to be clustered in ways that can isolate any specific phenomena to be modelled with the appropriate physical model. Using clustering algorithms can make the models more translatable to different problems with different kinds of sensors.

3.2.4 Score for collision

In order for a machine learning model to learn a behaviour, the expected correct behaviour needs to be clear and measurable. Generally, this is the label of a data input, or the output expected after the input is processed by the network. Simply using "1" for

the segment that causes the collision and "0" otherwise does not differentiate between different segments in a frame. In that case, a score formula was made to give an approximation of the danger in each frame by each segment.

$$A = \frac{f^2}{F^2} \cdot \frac{s}{S} \cdot \left(1 - \sqrt{\sum_{i=1}^2 (\mathbf{c}_i - \mathbf{C}_i)^2} \right)$$

The formula consists of three parts, and each one varies from zero to one, where one means danger and zero means safety. All of them combined give the final score A . The first element determines how close in time the current frame f is to the last one F , which can indicate a collision. The numbers are squared to put a bigger emphasis on the later frames than the early ones because these are the ones that are close to the collision. Second is how big an area the segment s occupies on the frame in comparison to the whole frame S . This is because from time to contact it is established that large objects in the image plane are likely close to the camera. Third is how far away the centre of the segment $\mathbf{c}$ is from the one of the frames $\mathbf{C}$. This distance is measured in the two coordinates of the point. Here, the coordinates are scaled so that the biggest possible difference is one. This is because a colliding object will, in general, be in the centre of the image plane, where it can hardly miss the UAV if too close.

As part of the improvements on the original experiments, an improved scoring function was also developed. This new score used a manually selected colliding object and the score was scaled by fifty per cent for each frame for that segment.

3.3 Data modeling

3.3.1 Hamiltonian neural network

To be able to understand the evolution of a dynamical system, there is a need for a discrete model to describe its behaviour at each time step. One such approach is to use Hamiltonian mechanics. This type of mechanics uses a function called the Hamiltonian to derive the state of the system from partial derivatives of the function at a specific time. The problem arises when there is a need to find the Hamiltonian of a system that is not explicitly known. In practice, this arises often because a lot of the time, scientists have to infer the structure of a system based on observational data. Here comes that Hamiltonian neural network described in section 2.2.1, which allows to estimate the Hamiltonian from observational data. The way that this is done is by adding a loss function that takes the already known generalised coordinates $\mathbf{q}$ and conjugate momentum $\mathbf{p}$ and their time derivatives as an input. Then the loss function tries to minimise the difference between the real-time derivatives and the ones gained from using the corresponding coordinates, momentum and current Hamiltonian. This way, the model learns to estimate time derivatives that align with the real world.

$$\mathcal{L} = \frac{1}{N} \sum_{i=1}^N \left[\left(\dot{q}_i - \frac{\partial H}{\partial p_i} \right)^2 + \left(\dot{p}_i + \frac{\partial H}{\partial q_i} \right)^2 \right], \quad (8)$$

Here $\mathcal{L}$ represent the loss for the particular prediction, where N is the number of dimensions of the output layer.

Because this is the first time ever optical flow data is used for a Hamiltonian neural network for this specific project, some design choices were made to select appropriate generalised coordinates $\mathbf{q}$ and conjugate momentum $\mathbf{p}$ and their time derivatives. The forces the Hamiltonian network is expecting to model are trajectories of objects. At every time step, the object has a position and a path travelled from the last time step that was estimated by the optical flow. So it is obvious to take the x and y coordinates in the image plane as the generalised coordinates $\mathbf{q}$, because they perfectly show the exact position of the object in time. For the conjugate momentum $\mathbf{p}$, the travelled distance is the direction and magnitude of the motion. In physics, generally, the momentum is equal to mass times velocity, but in this thesis's case, the mass is excluded from the equation because it is not possible to estimate. This leaves the velocity mathematically equal to the momentum. This exclusion removes important information from the understanding of the scene, but in this specific context, this loss is not significant. This is because, in the field of collision detection, most of the time incoming obstacles are considered in a small time frame of their flight. This makes the trajectories relatively constant over the time of interest, so the precise impact of the gravity on the obstacle's trajectory is not needed. Furthermore, static or ground-based obstacles are generally not affected by the loss of mass measurement since, in these cases, gravity does not affect their motion that much. This is useful because the time derivative with respect to the coordinates gives the velocity of the motion. In this case, the same optical flow measurement for this time derivative can be used, too. In general, the velocity examines the change in coordinates in time, so it makes sense to be the distance from the previous time step. The last parameter is the time derivative with respect to the momentum. This in physics is generally represented by the force that acts on the system, or in other words, how the velocity changes over time. Again, there is no way to estimate the force of gravity or drag through the air, but there is one force that can be measured, which is acceleration. This is because with neglected mass from Newton's second law of motion $F = m \cdot a$, the only aspect of the force acting on the object is the acceleration. This can be measured by the difference in the previous two velocities.

After all the measurements are established, the transition between time steps needs to be described. The method for transition between states is time forward integration. It is used to compute velocities and positions from the acceleration. For the update of the velocity in each time step, the corresponding acceleration to the previous velocity is added.

$$v_{n+1} = v_n + a_n$$

Here v_n is the velocity of the current time step and v_{n+1} of the next one, and a_n is the acceleration.

On the other hand, the update of the coordinates adds not only the velocity but also half of the acceleration, so the model will be more sensitive to fast-accelerating objects.

$$c_{n+1} = c_n + v_n + \frac{a_n}{2}$$

Here c_n is the coordinates of the current time step and c_{n+1} of the next one. a_n is the acceleration and v_n is the velocity of the current time step.

One additional improvement of the model that was tested is the use of a more complex structure of the motion to be learned by the Hamiltonian neural network. In general,

neural networks are quite good at modelling high-dimensional vectors, so it was useful to try this aspect here too. So instead of modelling the motion of one of the nine points at a time, all of the nine coordinates and momenta were given as an input to the network. This means that the motion was categorised by a thirty-six-dimensional vector. This dynamical system is almost impossible to model by a human with all this complexity, but it is absolutely possible to estimate with a neural network.

To sum everything up, with these design choices, all elements of the Hamiltonian structure of motion were covered only with optical flow measurements. With this structure, predictions from three frames in the future can be made about the motion of the observed object.

With all said in section 2.1.1, Kalman filter looks perfect for this project. Here, there is noisy measurement data, which is the optical flow and possibly a noisy model of the motion, which is the Hamiltonian neural network. In that case, being able to tell when to trust these measurements will provide a much smoother and accurate representation of the motion.

3.4 Conclusion

To sum up the methodology of this thesis, all of the design choices of this project were taken into account in discovering a new way to interpret concrete physical measurements about the real world. The choice of data and its preparation were done to gather the most physical insights about the motion of the object in the image plane. After that, the modelling was guided by the philosophy that this is the first time this kind of data has been modelled in such a way, which led to several logical assumptions about the design. In the next chapter, the results from experiments to examine the effectiveness of the proposed design will be presented.

4 Results and discussion

In this section, the results from all performed experiments will be examined. After describing the experiment and the result, a possible explanation for the cause will be developed. First, the architecture of the proposed models and experiments will be described in section 4.1. After that, the results from training the Hamiltonian neural networks will be analysed in section 4.2. Furthermore, information about the performance of all Experiments on both types of scoring functions will be presented in terms of accuracy and time for training the model in sections 4.3 and 4.4. Following that, an interesting case for complex motion will be described in section 4.6. And finally, limitations of the design and their mitigation will be explored in section 4.7.

4.1 Validation

The main propositions of this thesis are whether there is a physically accurate way to model visual data(RQ1 1) and whether this can be used in the field of UAV collision detection(RQ2 1). With all the gathered data, there can be supported evidence for these propositions within the constraints of the research project. Several experiments were conducted, each to test a specific aspect of the system and were validated based on their performance.

This section describes all experiments that were used to measure the performance of the proposed collision detection method. Each experiment was meant to show a different aspect of the method. The first experiment was the proposed method that includes HNN prediction of three steps in the future of the original optical flow data. For the final prediction about whether the particular frame indicates a collision, a feed-forward neural network was constructed to process the information. For the particular choice of hyperparameters, several iterations were done to optimise them. This is the architecture of the best-performing model (figure 13).

```

model = Sequential([
    Dense(units: 256, activation='relu', input_shape=(76,)),
    BatchNormalization(),
    Dropout(0.2),
    Dense(units: 328, activation='relu'),
    BatchNormalization(),
    Dropout(0.2),
    Dense(units: 328, activation='relu'),
    BatchNormalization(),
    Dropout(0.2),
    Dense(units: 328, activation='relu'),
    BatchNormalization(),
    Dropout(0.2),
    Dense(units: 116, activation='relu'),
    BatchNormalization(),
    Dropout(0.2),
    Dense(units: 1, activation='sigmoid')
])

```

Figure 13: This is the feedforward network used for the collision detection prediction. It has one input layer and four hidden layers, all with a rectified linear activation function. To combat overfitting, batch normalisation and dropout with a rate of 0.2 were used between each layer. The output layer has a sigmoid activation function.

To determine whether the additional data points are beneficial, an experiment without them was conducted. Next, to determine if only considering the number of frames the segment was present is enough to grasp the time aspect of the problem, another Experiment with the HNN additional data, but using LSTM with 64 units to look at ten frames sequentially. Next, to determine whether the optical flow needed to describe the movement of the object, an RNN without any optical flow data was tested. This model had information only about the size and position of the segment in the image plane. The RNN has two hidden layers with sixty-four neurons each. The main difference of this model compared to the feedforward network of the first experiment is that it processes ten frames sequentially at one iteration and calculation of the loss, not just one. At the end, a standard CNN model was compared because it is considered a standard method to work with image data. As for the construction of the CNN, it has two convolution layers with sixty-four and thirty-two masks. Both kernel sizes are three with a padding of one. Furthermore, there is a pooling layer with a kernel size of two and a stride of two. It then finishes with two fully connected layers. Next, Experiments were prepared to test the improvements of looking at the whole flow, using a Kalman filter and clustering the flow before the model. All three improvements are tested on the architecture of the first experiment with the HNN predictions. Finally, all of the aforementioned models were tested a second time on the improved scoring function as a label.

The metrics used for evaluating the model are mean average error(MAE) and time to train the model on seventy-five epochs. The MAE is needed to evaluate how close the predictions are to the chance of collision with the proposed score. The time to train is an important metric because it determines how quickly the model can respond to new

information.

4.2 HNN Training Loss on Optical Flow Data

The first results that need to be discussed are the training loss of the HNN models on optical flow measurements gathered from the main dataset. This is measured in terms of mean average loss.

Method	Loss
HNN One Point	0.0441
HNN Full Flow	0.062067
HNN Clustered Flow	0.066865

Table 3: HNN training loss results on optical flow data

The first question that has to be discussed is whether this formulation of the Hamiltonian neural network is able to model such sparse and noisy data as the one collected from optical flow. By examining the final training loss on each model in Table 3, it is observed that it is less than 0.1. This indicates a quite close relation between the prediction of the model and real coordinates and velocities. This can be interpreted by the drastic reduction of the complexity of the real-world motion to the essential observable components of position, velocity and acceleration. In the literature [25], tests on Hamiltonian neural networks are done on dynamical systems with well-defined physical parameters. From single to double pendulum, these are isolated experiments with hard but possible to infer states in time. On the other hand, in this example, the motion examined has no ability to be calculated but only estimated. This is due to the fact that the parameters used, like position, velocity and acceleration, are 2D projections of the real-world motion. They indeed have a physical backing, but can not be predicted by any ODE because there are many unknown parameters like height, mass or drag. This can also be seen later in the difference between the experiment with and without the HNN data. They are quite close in their predictions and time, which on one hand can be interpreted as an insufficient result in the collision detection sense. On the other hand, their close results show the similarity of the motion examined by optical flow and modelled from the HNN. This adds to the gap in the research of physics-informed machine learning applications in implied motion. Without a full descriptive picture of all forces in action being measured, the approximated motion closely resembles the sensory one. This is, in general, the idea of a neural network to be able to estimate solutions even from limited data, since if all the necessary information were present, ODEs would have been better.

4.3 Results with Original Score Algorithm

Experiment	Mean Average Error	Time to Train (s)
With HNN Experiment	0.0216	33.36
Without HNN Experiment	0.0195	33.02
LSTM	0.1356	40.57
RNN	0.0595	106.19
CNN	0.0153	668.92
Full HNN Flow	0.0191	34.17
Clustered Flow	0.0223	32.94
Kalman Filter	0.0937	35.44

Table 4: Performance comparison using original score algorithm

In this section, the results of the first scoring function described in section 3.2.4 will be examined. One hypothesis that was proposed in this project was that when designing models that take into account relatively constant motion, there is no need to look at several consecutive instances of this motion, but to simply consider the amount of time the motion was present as described in section 1.4. The result of this can be observed between the experiment using LSTM and the base experiment, where HNN data was used, which are rows one and three in table 4. From the results, it seems that the average error is three times larger in the LSTM example. This can be explained by trying to explain a simple relation with an excessively complicated model. This approach can be useful in complex examples like learning optical flow from particle image velocimetry data [35] or where many different features extracted from CNN have to be modelled in time [43]. Otherwise, in relatively constant motion, this overhead does not provide any additional benefits. With this taken into account, future improvements in the collision detection system can be made to simplify stable motions for faster, cheaper and more accurate predictions.

Looking at the second research question proposed by the thesis (RQ2 1), there is a need to examine how different approaches for collision detection compare to one another. On the first iteration of the scoring algorithm, it is examined that optical flow with and without the HNN data performs relatively equally, which can be seen in the first two rows of table 4. This can be explained by the close predictions of the HNN model, which, even taking into account three steps into the future, do not provide enough relevant data to improve the performance. Next are the more complex models of LSTM with HNN data and RNN without any optical flow data, which are lines three and four of table 4. From the above paragraph, LSTM brings too many excessive computations to the problem, which affects the error. For the RNN that was modelled closely like a time-to-contact system, the low accuracy shows the need for optical flow. The explanation here is that this kind of model does not raise an alarm until there is a noticeable difference in the size of the incoming obstacle, so slow-moving or small in general ones give a false sense of security for the time. Next comes the CNN, which is the model most likely to be used in any computer vision application, for example [43], which is on the fifth line of table 4. This is proven to be right because it gave the best performance overall. The downside here is that the convolution algorithm is very computationally heavy, and this is reflected

in the time needed to train the algorithm, which is twelve times longer than the first experiment. This shows that here accuracy is traded for time to response, which in the context of collision detection, where decisions need to be made in fractions of a second, is a disadvantage. This confirms the need to use motion information for better predictions of collision detection, similar to [40] or [42]

4.4 Results with Improved Score Algorithm

Experiment	Mean Average Error	Time to Train (s)
With HNN Experiment	0.0174	60.28
Without HNN Experiment	0.0124	70.69
LSTM	0.0549	62.36
RNN	0.0294	60.73
CNN	0.0036	2819.15
Full HNN Flow	0.0122	88.77
Clustered Flow	0.0119	124.25
Kalman Filter	0.2775	83.18

Table 5: Performance comparison using improved score algorithm

After the original set of experiments was conducted, several possible improvements were implemented to better tackle the problem. The first one was changing the score that represents a danger as explained in section 3.2.4. To do this, after manually picking the crashing segment, all of the scores for this segment were scaled up. The logic here is that in the examples where an actual danger is present, the danger needs to be spotted earlier, before the size of the incoming obstacle grows. This, in theory, has to rely more on the HNN predictions than on the size and centerness of the obstacle. The result was reducing the error in all of the models, but unfortunately, not proportionally to the ones that use HNN data. This can be observed if comparing all rows from table 4 to table 5 respectively.

The next improvement that was proposed was to use a more complex Hamiltonian neural network to capture the motion. Here, the network was trained on the input containing the optical flow measurements from all nine points of the grid. The idea for this is that complex motions like rotation or “focus of expansion” can be explained better if there is motion in different locations simultaneously. This proved to be successful, with this being the best-performing model after the CNN in both types of scores. This can be seen in the sixth row of each table 4, 5.

The next proposed improvement was to cluster the optical flow instead of using a grid structure to have multiple different measurements of the flow from one segment, as mentioned in section 3.2.3. In theory, the more flexible structure of the clusters will produce more natural flow estimation and lead to better results. This was also confirmed on the improved score experiment, where it was just below the full flow experiment. This can be seen in the seventh row of each table 4, 5.

Logical improvement of the flow was the use of the Kalman filter as discussed in section 3.3.1 [17]. The justification for this comes from the fact that this system is represented by noisy measurements that are the optical flow and model of the motion in the HNN.

In that case, the Kalman filter has to be useful for smoothing the expected trajectory. Unfortunately, this was proven quite ineffective, but this can be due to the use of a common Kalman filter not suited for this example. This can be seen in the eighth row of each table 4, 5.

4.5 Visual example

In this section, a visual example of the system will be shown. The example presented is a bird flying through the UAV and crashing into it. First the original segmented frames will be displayed in figure 14. They will be followed by ordinary optical flow measurements in figure 15, and finally, the HNN predictions will be displayed in figure 16.



Figure 14: Original images.



(a) Time step 1



(b) Time step 2



(c) Time step 3



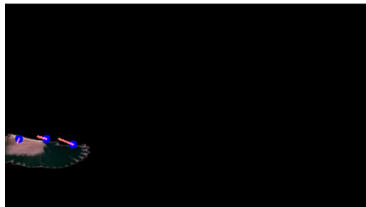
(d) Time step 4



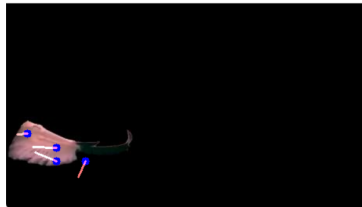
(e) Time step 5



(f) Time step 6



(g) Time step 7



(h) Time step 8



(i) Time step 9

Figure 15: Optical flow measurements.

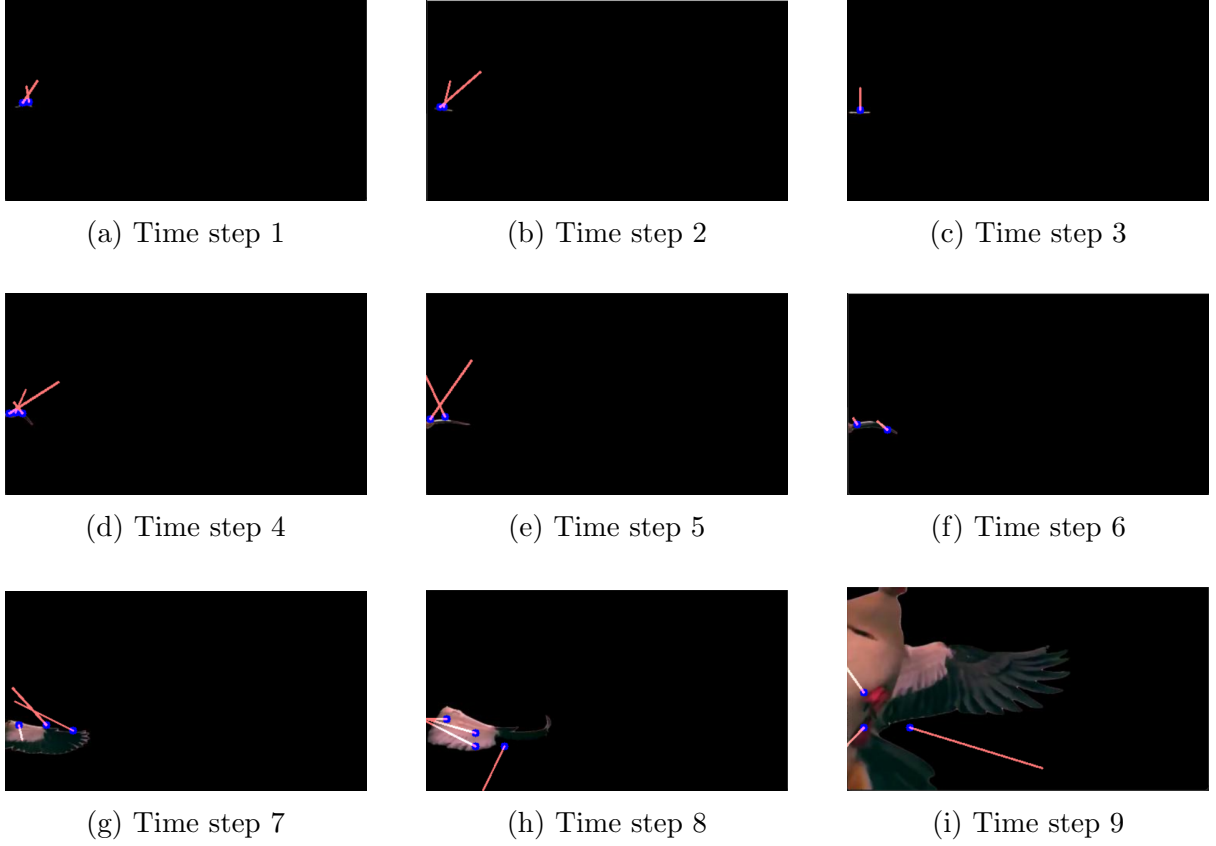


Figure 16: HNN predictions.

From all of the displayed images, it can be shown how, in the HNN examples, the motion is visible much earlier, so a manoeuvre can be taken before the object is too close. Because the HNN is making predictions for three steps in the future the acceleration makes the motion much more visible and consistent between frames.

4.6 Rotational motion from optical flow

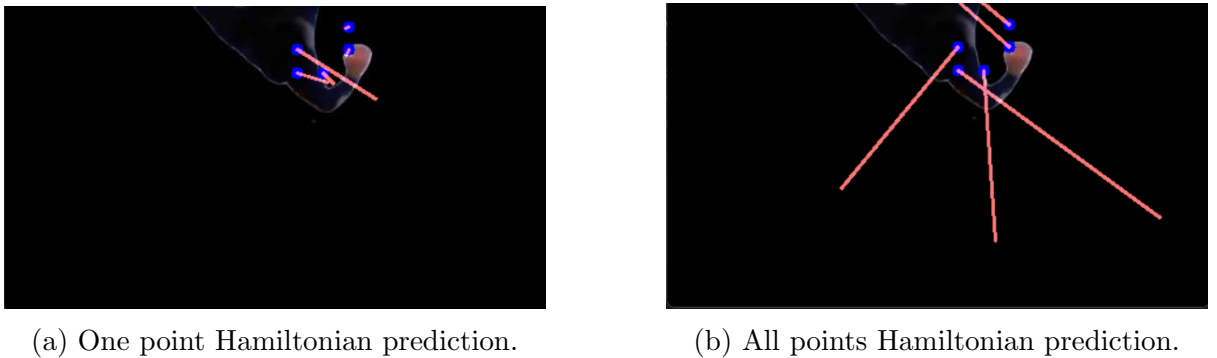


Figure 17: This is a comparison of rotational motion modelled with the two proposed Hamiltonian neural networks. In the images, there is one frame of a person doing a backflip in the air. The red lines are optical flow predictions from the two HNN models.

Finally, some insights were gathered from the optical flow calculated between the two Hamiltonian neural networks (Figure 17). In the dataset, there is one example with a

more complex type of motion. The video consists of a person on a skateboard who makes a backflip and hits the UAV. This is a three-hundred-and-sixty-degree rotational motion. This is an interesting example because this type of motion is much more complex than other trajectories in the dataset. From the visualisation of the flow, it is clear that the network modelling all the optical flow points at the same time was able to describe the motion much more accurately. This is characterised by optical flow measurements in all directions describing a circular motion. From this observation, it is clear that the model with all points has a much better understanding of complex motions, and this is translated to the score on the collision detection experiment.

4.7 Limitations

There are several known limitations of many components of the system. Nevertheless, all the possible steps were taken to mitigate their effect on the performance of the system.

4.7.1 Data limitations

Starting from the data, the first limitation is that the correct trajectory of an obstacle is not present in the data to make a comparison. This means that the accurate real-world trajectory cannot be examined in comparison with the one predicted by the Hamiltonian network. With all this in mind, the only way to predict the accuracy of the physical model is by the indirect effect of an accurate trajectory. The way this issue is tackled is by comparing the accuracy of collision detection with and without the additional physical data. The second issue with the data is the missing annotation about which segment of the image causes the collision in the last frame. This was handled with a custom scoring function to predict the level of danger proposed by a particular segment at any time steep.

4.7.2 Segmentation limitations

On the side of the video segmentation, the algorithm performed badly in crowded scenes with several similar objects. Another problem was the prolonged occlusion, where one object stayed hidden for a long time. And the final limitation, which was especially prevalent in this situation, was missed details for fast-moving objects. For the first one, a more precise mask was chosen to distinguish between similar objects. For the second one, in case of examples with long occlusion, manual stitching of the frames was done. And the third issue was not such a serious problem in this specific project because there were enough optical flow measurements, even if some details were missing, to describe the flow.

4.7.3 Optical flow limitations

For the well-known limitation of Lucas Kanade optical flow about larger motions, different from the local ones, were fortunately not present due to the particular dataset, but have to be kept in mind with additional data. The sensitivity to noise is another common limitation which was prevalent but was mitigated by averaging many measurements at the grid and by outlier cleaning of the data. The most prevalent limitation is the brightness constraint or slow-moving objects, which lead to missing measurements in entire frames.

This was mitigated by adding partial optical flow from previous consecutive frames, so there will not be empty measurements.

4.7.4 Lable limitations

The custom score for the collision was a limiting factor for the model accuracy, but this was mitigated by scaling it with the actual danger represented by the particular segment.

4.7.5 HNN limitations

The biggest limitations by far are the ones around the Hamiltonian neural network. This is mainly because this model was not meant to be used on such low-fidelity and sparse data. Usually, these models require a known physical model to be examined. Furthermore, the rigid physical structure of the architecture made it underperform in more subtle and fast motions. This is because if the conservation of energy is not well defined in the data, then the biggest strength of the Hamiltonian neural network becomes a burden. Furthermore, in physics-informed machine learning, there is a common limitation that the design and coordinate system differ widely from example to example. With taking all of this into accaunt, there were several design choices meant to mitigate these effects. The most important one is the great simplification of the motion. Forces like gravity or drag and parameters like mass and aerodynamics were omitted, so the only variables observed were position, velocity and acceleration. This simplicity can easily satisfy the rigid physical constraints of energy conservation. At the same time, the momentum needs to be conserved because in many cases, the acceleration of the obstacle is relatively constant throughout the videos. At the same time, this is a simple but effective design and coordinate system that can be relocated to new data or many similar problems.

4.7.6 Real world limitations

The main limitation of this thesis is limited experience in real-life situations, like closely avoided crashes, which can decrease the effectiveness of the algorithm in these kinds of obstacles. Of course, real-life data for crashing drones will be very expensive to obtain, but some controlled environment should be considered and even some tests on a physical UAV. Nevertheless, even without certain situations and real-life tests, the system can signal danger of collision in conditions very similar to the real world.

5 Conclusion

This thesis aimed to be a blueprint for modelling data from visual sensors with a physically backed machine learning model (RQ1 1). And to prove the effectiveness of the models, they were used in a real-life problem of UAV collision detection(RQ2 1).

This project develops a better understanding of how the fields of computer vision and physics-informed machine learning can support each other. By examining ways to represent noisy and sparse computer vision data into a simple latent space, a rigid physical model can be trained on this representation. These models can bridge the gap between low-cost sensors and physically accurate and interpretable models. Furthermore, this thesis examines the physical understanding of the real world on the problem of collision detection, where it was proven to be an effective consideration. This can benefit any kind of low-cost UAV businesses to have safer operations.

5.1 Recommendations for future work

For future developments in this field, it has to include a simulated environment that can train the model in more situations and diverse conditions. Situation like close passing or rotating objects are not present in standard datasets, but can easily be simulated. This can strengthen the ability of the UAV to handle different situations similar to those expected in the real world. From the physics-informed computer vision perspective, there should be more research on modelling motion from computer vision with physics-informed machine learning. Complex motions like rotations and pivoting, or even precise modelling of a trajectory, can all be modelled in a way to satisfy physical laws in the real world for more interpretable and reliable results.

5.2 Take away

The key takeaways from the findings of this thesis are that sparse, noisy or low-fidelity motion from simple sensors still can be modelled in a way to satisfy physical constraints. Other than that, this kind of information is key to better interpreting the trajectories of incoming obstacles through UAVs. With the lack of standard methods of modelling visual data like optical flow in a physically accurate way, this project can be used as a guideline on how to grasp all the necessary aspects of a physical motion from limited sensory data. Additionally, this provided an unsupervised way to train the model without the need for human annotation. This made the system quite scalable with the newly available data, and it can be used in many different engineering problems.

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